

Key Points:

- Mixed-methods study combining bibliometric analysis and a field-wide survey to assess Earth science engagement in environmental justice (EJ)
- Bibliometric results show Earth sciences rank seventh in EJ scholarship, lagging behind environmental and social sciences
- Survey findings indicate career stage and race shape knowledge, community engagement, and perceived funding access among Earth scientists

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Where on Earth Are the Earth Sciences? A Mixed-Methods Study of Environmental Justice Research Trends and Roadblocks to Engagement

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Abstract Environmental justice (EJ) research relies on expertise from multiple disciplines to address community-specific health risks and inequities. EJ is an inherently transdisciplinary framework, yet the expansion of EJ scholarship has not occurred evenly across academic disciplines. Understanding disciplinary contributions is essential for identifying gaps in knowledge and practice. This research employed a mixed-method approach to investigate EJ research trends and roadblocks to engagement. Study 1 used a bibliometric analysis of Scopus-indexed publications and found that Environmental science and Social sciences were the leading contributors to EJ scholarship. Earth sciences ranked only seventh in EJ research volume by proportion and exhibited comparatively stagnant growth. Study 2 investigated this discrepancy through a survey of Earth scientists ($N = 460$) examining perceived barriers to EJ engagement. Results indicated that career stage was a significant predictor of EJ familiarity and community involvement: early-career Earth scientists were more familiar with EJ but were less involved in their communities' environmental issues than their late-career counterparts. Race and ethnicity were also associated with differences in perceived funding difficulty, with White respondents reporting fewer challenges in securing support for EJ-related projects. Taken together, these studies assess EJ engagement within the Earth sciences and identify structural and experiential factors that may limit participation. Identifying disciplinary gaps and roadblocks helps integrate Earth sciences into EJ scholarship and increase research on environmental health and equity.

Plain Language Summary Environmental justice (EJ) is the belief that everyone deserves to live in a safe, healthy environment. To help make this possible, researchers from different fields work to solve problems related to pollution, health, and fairness. In this study, we analyzed thousands of EJ-related research papers in Scopus, a large academic database. We found that Social and Environmental sciences produce the most EJ research and are growing rapidly in this area. Earth sciences, despite being highly relevant to environmental topics, ranked only seventh in EJ scholarship and showed slower growth. To understand why, we conducted a national survey of Earth scientists. We found that early-career scientists were more familiar with EJ ideas but were less active in local environmental efforts. In contrast, more senior Earth scientists were more engaged with communities but less familiar with EJ. We also found racial and ethnic differences in how people viewed funding access: White scientists were more likely to report that funding for EJ work was easier to obtain. These findings suggest that better support systems and inclusive training may help Earth scientists engage more in EJ. Their unique expertise could bring new ideas to help protect communities and the environment alike.

1. Introduction

The environmental justice (EJ) movement has long highlighted the disproportionate environmental burdens borne by marginalized communities, linking environmental degradation to structural inequalities in health, safety, and quality of life (Bullard, 1993; Cutter, 1995; Pastor et al., 2001). From hazardous waste siting to unequal exposure to pollutants, EJ has become a critical framework for understanding how environmental and health disparities intersect (Burger & Gochfeld, 2011; Lanphear et al., 1996).

The integration of EJ into the health sciences has been naturally driven by the recognition that individual health outcomes are shaped by environmental and socioeconomic factors (Brulle & Pellow, 2006; Mohai et al., 2009). Landmark reports and events in the 1980s and 1990s, including the 1987 United Church of Christ report on "Toxic Waste and Race" (United Church of Christ Commission for Racial Justice, 1987) and the 1991 First National

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Peoples of Color Environmental Leadership Summit (Alston, 2010), have centered and highlighted the disproportionate environmental health stressors on minority and under-served communities in the United States (Bowen & Wells, 2002). However, national health policy initiatives in the United States have yet to see a complete embrace of EJ, despite early appetite driven by the 1994 Executive order directing federal agencies to develop EJ considerations in their programming and function (Buckhoy, 2015; Section et al., 1999). Instead, a number of health studies have driven the research and framework development of EJ to provide insight on how diverse communities are impacted nationwide (Casey et al., 2023; Garcia et al., 2013; Shepard, 2002).

EJ is an inherently transdisciplinary framework, enabling broad fields such as the social sciences, business, and public health to integrate environmental, societal, and health-based expertise in addressing inequality (Ernstson, 2013; Ramirez-Andreotta et al., 2016; Reynolds & Ciplet, 2023). However, the expansion of EJ scholarship has not occurred evenly across academic disciplines. While some work has mapped the broader growth of EJ research, few studies have systematically disaggregated scholarly contributions by field (Mohai & Saha, 2015; Temper et al., 2015). This limits our ability to identify where disciplinary gaps persist and where untapped potential for advancing EJ remains.

One relevant field that has yet to measure EJ scholarship is the Earth sciences. The Earth sciences broadly encompass the physical and environmental sciences, providing critical insight on how global feedback systems function and evolve on human and geologic timescales (Von Engelhardt & Zimmermann, 1988). Expertise contribution from the Earth sciences would reveal novel paths for research and collaboration in EJ; however, the field is small and lacks diversity compared to other larger STEM fields, revealing a possible roadblock toward broad EJ engagement (Bernard & Cooperdock, 2018).

This paper aims to address how different academic disciplines, particularly the Earth sciences, contribute to EJ scholarship—an area with implications to public and environmental health outcomes. Study 1 uses a bibliometric approach to quantify the disciplinary distribution of EJ publications over a 30-year period, highlighting the Earth sciences as a field whose contribution lags behind despite its central relevance to environmental inequities and health disparities. Study 2 builds on these findings by exploring possible reasons for this engagement gap. Drawing on a national survey of Earth scientists, it examines researchers' perceptions of funding access, difficulty of inclusion, and knowledge related to EJ, factors that can limit their participation in EJ work. Together, the two studies offer an initial empirical basis for identifying the structural and perceptual barriers that may constrain EJ engagement in the Earth sciences.

2. Study 1: A Bibliometric Approach

2.1. Introduction

The rise in public interest toward EJ has been led primarily by grassroots movements in efforts to bring attention to the socio-economic and environmental inequities that exist within various communities (Bullard & Johnson, 2000). These movements have found some success in influencing health policy in the US (Chowkwanyun, 2023). However, its impact on academia has yet to be fully understood due to the complicated nature of the topic and the lack of dedicated scholars in institutional spaces (Carlos Garibay et al., 2016; Rabe, 2024).

Bibliometric analysis provides a systematic method for assessing scholarly impact by identifying publication trends, citation patterns, and topical evolution within academic fields. Study 1 tracks the growth and visibility of EJ-related work through Scopus. The purpose of Study 1 is to determine the extent to which Earth scientists have contributed meaningfully to EJ scholarship, as evidenced by research publications that explicitly employ EJ terminology.

2.2. Materials and Methods

2.2.1. Software and Tools Used

This study utilized the Scopus Database to conduct a bibliometric analysis of the absolute number of EJ-related publications and their growth trends over time, leveraging the platform's advanced search and categorical filtering features (<https://www.scopus.com/search/form.uri?display=basic#basic>). A two-step approach was used to identify and analyze a target population of EJ articles. First, an adapted Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) method (Moher et al., 2009) was developed to structure search criteria

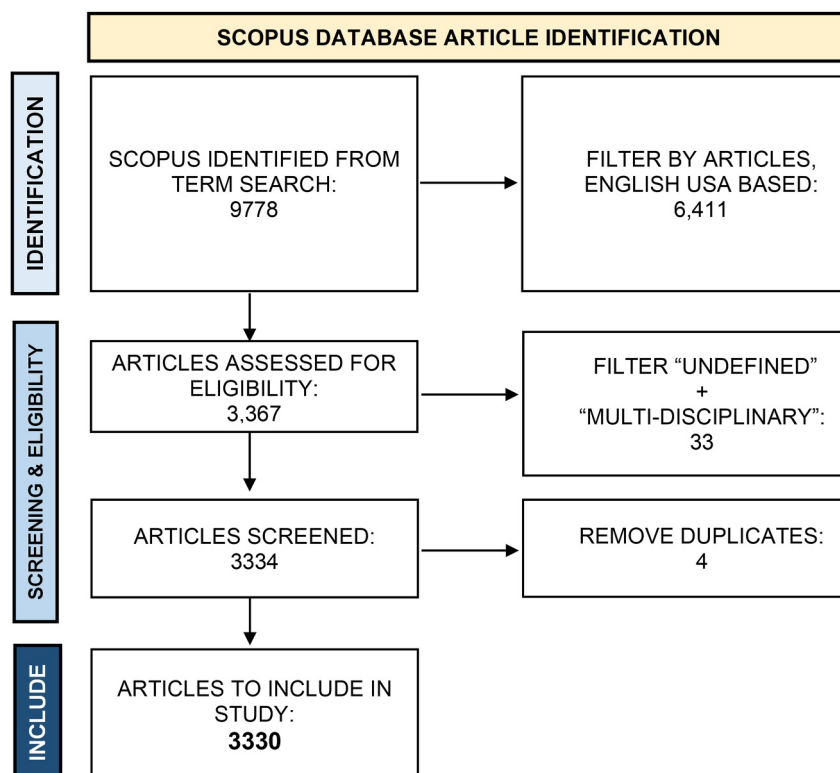


Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart outlining the identification, screening, and eligibility process for the article numbers to be included in this study. A total of 6,448 items were removed from the initial search to develop a robust and representative population of peer-reviewed research articles.

and identify a representative population of articles for analysis (Figure 1). The Selected population was then analyzed for trends in Excel and Python through Jupyter Notebook.

2.2.2. Scopus Literature Search

The Scopus database was chosen for this study due to its broad coverage of peer-reviewed academic literature and its robust “subject area” filtering feature, which is ideal for examining disciplinary patterns and gauging academic interest in EJ across a wide range of research fields (Ferretti et al., 2016; Hassan et al., 2014; U.S. Department of Education, 2019). Through its advanced search tools, the “TITLE-ABS-KEY” was used as the foundation for this study's inquiry to search the database for all peer-reviewed articles that mention or have a tag of “EJ” or “climate justice” in article metadata for every year from 1991 up until 2022. This initial screening returned 9,778 items. Additional filters were then included to account for subject areas, article type, region and publication language, which produced the following final advanced search strings used:

Final “EJ Related” articles to include:

TITLE-ABS-KEY (“EJ” OR “Climate Justice”) AND PUBYEAR <2023 AND (LIMIT-TO (SRCTYPE, “j”)) AND (LIMIT-TO (AFFILCOUNTRY, “United States”)) AND (EXCLUDE (SUBJAREA, “Undefined”) OR EXCLUDE (SUBJAREA, “MULT”)) AND (LIMIT-TO (DOCTYPE, “ar”) OR LIMIT-TO (DOCTYPE, “English”)) AND (LIMIT-TO (LANGUAGE, “English”))

An “all article” population that does not include keywords over the same filters:

PUBYEAR <2023 AND (LIMIT-TO (SRCTYPE, “j”)) AND (LIMIT-TO (AFFILCOUNTRY, “United States”)) AND (EXCLUDE (SUBJAREA, “Undefined”) OR EXCLUDE (SUBJAREA, “MULT”)) AND (LIMIT-TO (DOCTYPE, “ar”) AND (LIMIT-TO (LANGUAGE, “English”)).

The search returned 3,334 EJ articles for further screening and an accompanying unfiltered “all article” population of 10, 264, 454 articles, using the same filters without keyword inputs as a control.

2.2.3. Selection and Variable Identification

The criteria for article selection were based on relevant terminology, publication time-frame, and the structural limitations of the Scopus database. Given that this study focuses on the academic visibility and diffusion of EJ as a concept, the inclusion of “EJ” as a primary search term served as a foundational component of the bibliometric search strategy.

While “EJ” is the primary term for this study, terms related to EJ, such as “climate justice” and “energy justice,” have emerged as distinct areas of scholarship with overlapping concerns (Jenkins et al., 2018; McCauley & Heffron, 2018). To assess whether relying solely on “EJ” would exclude relevant literature, separate searches were conducted using these additional terms. The term “energy justice” yielded no substantial number of unique articles beyond those already captured in the initial EJ search. In contrast, “climate justice” returned approximately 150 unique articles not included under the EJ search. Given the conceptual and practical links between climate justice and EJ (Schlosberg & Collins, 2014), both “climate justice” and “energy justice” were included in the final search strategy to ensure broader and more representative coverage of the literature.

The chosen time frame for this study was informed by both the Scopus database's historic coverage limitations and the aim to account for the earliest identified publication trend relevant to EJ. While one article in 1986 was identified to be the earliest record of EJ academic literature in the Scopus database (Dubé, 1986), annual publication consistency did not begin until 1992. For this reason, the trend analysis began in 1992, which also coincided with a major milestone in the EJ movement—the First National People of Color Environmental Leadership Summit in 1991 (United Church of Christ Commission for Racial Justice, 1987). The endpoint chosen was the most recently completed publication year from this study's inception, 2022. This defined a 30-year period of observation for trends in EJ academic literature.

The filtering criteria were also influenced by the database's reliability in its coverage. Although Scopus offers comprehensive coverage of peer-reviewed journal articles across scientific disciplines, its indexing of other publication types, such as conference proceedings and reviews, is less consistent (Pranckutė, 2021). For this reason, analysis was restricted to journal articles only.

While the EJ movement has global relevance, its academic development has been most extensively documented within the United States (McGurty, 2007; Perez et al., 2015). The geographic scope is therefore limited to U.S.-affiliated publications. Similarly, because Scopus coverage is most robust for English-language journals (Mongeon & Paul-Hus, 2016), the study focused exclusively on English-language articles to ensure consistency in data quality.

Although EJ is inherently multidisciplinary, Scopus articles labeled only as “Multidisciplinary” or “Undefined” were excluded from the analysis. This is because these categories were rarely used by publishers in the sample (only 33 articles), often overlap with already cross-listed subject tags, and did not contribute meaningfully to identifying disciplinary trends in EJ scholarship (Álvarez-Llorente et al., 2025).

The final screening technique removed duplicates, which were primarily caused by articles submitted and published in more than one journal presenting the same or similar results under one DOI/title. This removed a total of four articles in the EJ population. This screening, however, was not done in the “all-article” population due to the limitations of large-scale data acquisition from Scopus. The “all-article” analysis was not expected to be impacted due to the presence of duplicates because the aim is to focus solely on the growth trend as a comparison; in-depth analysis was only done with the “EJ Articles” population. The resulting data was then processed in Excel, with trend analysis being done in Jupyter Notebook Version 3.11.5.

2.3. Results

The “All Article” population followed a strong linear trend ($r^2 = 0.94$), while the EJ population followed a strong exponential trend ($r^2 = 0.96$) (Figure 2). EJ research remained on a rapid growth trajectory in academia, where 2022 saw a year-end output of over 440 articles, a 9% increase from 2021.

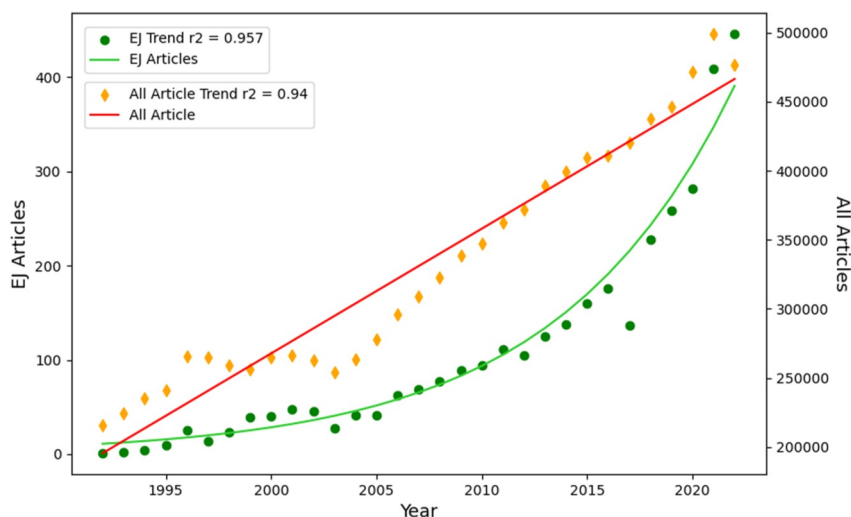


Figure 2. Article publication trends for environmental justice (EJ)-related articles and all articles per year from 1992 to 2022. This 30-year timespan demonstrates a linear growth ($r^2 = 0.94$) by the “all articles” group that was identified using Preferred Reporting Items for Systematic Reviews and Meta-Analyses filtering without using the EJ keywords. The “EJ articles” group demonstrates an exponential growth trend ($r^2 = 0.957$) for the same period.

Interestingly, the “All Article” population saw a decrease in 2022 publication productivity of 4.50% from the year prior. This was the first time since 2003 that general publication trends had seen a drop. However, evidence suggests that the past 2 years were abnormally high publication years due to the adjustments academics made to focus more on written work and journal submissions (Garcia et al., 2013). This post-pandemic drop, however, was not observed in the 2022 EJ population. The trend continued exponentially, rendering the post-pandemic productivity levels of EJ scholarship a deviation from general academia trends.

One noticeable deviation in the EJ publication trend was observed in 2017, where there was a 22% drop in EJ article publication. This was not reflected in the all-article population, which saw a modest 2.19% increase over the previous year. A second deviation was seen in 2021, where the EJ publications showed a 45% increase over the previous year, with the “All Article” population seeing only a slight 5.81% growth rate over the same period. The subsequent years of 2018 and 2022 returned to normalcy for the EJ populations, where growth rates returned to expected levels along the exponential trend.

A more detailed examination of article growth by discipline, using the “Subject Area” filter in Scopus, reveals which academic fields have contributed most to EJ scholarship (Figure 3). The majority of publications originate from the Environmental sciences and Social sciences, two domains where cross-listing is common. This overlap reflects the interdisciplinary nature of EJ research, which often requires integrating environmental and social perspectives to address complex, justice-related challenges (Maxwell et al., 2019). In contrast, contributions from other fields remain comparatively limited.

Some highly productive fields, like engineering, have published a respectable number of EJ-related papers. However, when normalized by the size of the field's overall publication output, their relative engagement with EJ is lower compared to less prolific fields like Business, Management and Accounting, or Arts and Humanities (Table 1).

Notably, Environmental science and Social sciences stand out as both highly productive and deeply engaged in EJ research, reinforcing their central role in interdisciplinary EJ scholarship.

2.4. Discussion

Study 1 employed a bibliometric approach and found an exponential trend in EJ publications from 1991 to 2022. This resonates with previous investigations into environmental health productivity, which have likewise highlighted the prominence of EJ case studies from 2000 to 2020 (Biswas et al., 2024; Zhuang et al., 2022). The results of Study 1 found a wide discipline-wise discrepancy: the top contributors were Environmental science and Social

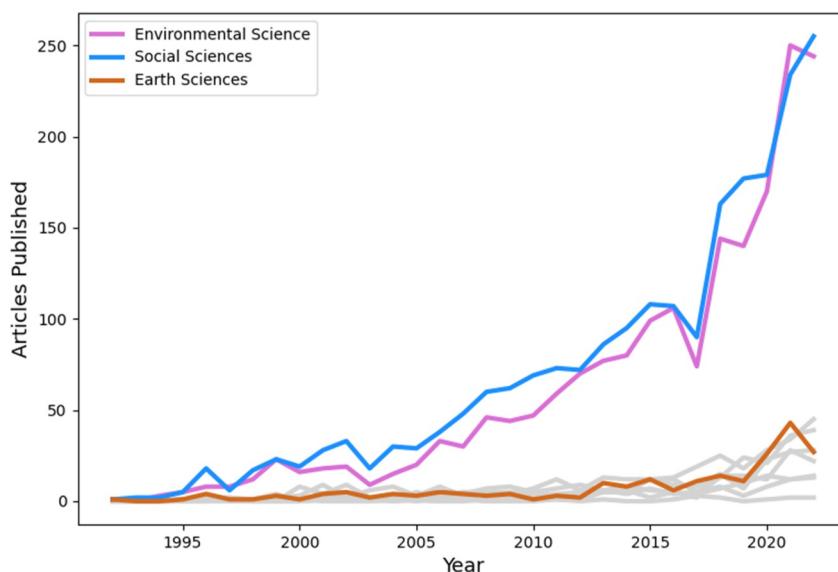


Figure 3. Total number of environmental justice (EJ) articles published by the top 10 contributing fields in Scopus. Environmental science and Social sciences were the top 2 contributors in the EJ space by sheer numbers, compared to Earth science, which lags behind all other fields.

science. Earth science, despite highly relevant to environment and equity, ranked only seventh in its EJ volume by proportion.

Two notable deviations occurred in 2017 and 2021. Because these were post-election years, there is reason to believe that political pressure and a shifting federal cabinet affected academia's funding process, especially with the increasingly polarizing topic of environmentalism (Birch, 2020). The subsequent rebound in production can be interpreted as a return to normal research operations after an initial period of uncertainty related to external election pressures. Recent federal policy changes suggest that EJ scholarship may face new political and funding constraints. EJ is among the words that federal agencies have discouraged using in grant-related contexts (PEN America, 2025). EJ researchers may need to avoid or rephrase EJ-related terms in grant proposals, which could make it harder to clearly describe the purpose and social relevance of their work. Funding opportunities explicitly tied to EJ may also become less available or less visible, reducing support for projects that study unequal environmental burdens. Furthermore, it may also make EJ research harder to track over time, as scholars may

Table 1
A Ranking of the Top 10 Fields' Contributions Toward Environmental Justice Research

Field	EJ articles	Total articles in field	EJ proportion (%)	Rank
Environmental Science (ENVI)	1,877	568,146	0.330	1
Social Science (SOC)	2,153	1,127,497	0.191	2
Energy (ENER)	185	186,695	0.099	3
Arts and Humanities (ARTS)	302	418,160	0.072	4
Business, Management and Accounting (BUSI)	155	292,706	0.053	5
Economics, Econometrics and Finance (ECON)	94	226,051	0.042	6
Earth and Planetary Sciences (EART)	220	557,908	0.039	7
Agricultural and Biological Sciences (AGRI)	159	844,687	0.019	8
Decision Sciences (DECI)	20	122,911	0.016	9
Engineering (ENGI)	164	1,025,502	0.016	10

Note. Fields are given an abbreviated code from Scopus, and their total raw publications are listed under the EJ article column. That is then averaged over their total publications to produce an "EJ Proportion" that makes up its final ranking.

avoid explicit EJ language in grant proposals, abstracts, and keywords even when the substance of the work remains closely related to EJ.

The overall exponential trend in EJ publications is distinct from the steady growth of the collective “All Article” publication group, emphasizing the desire and ample supply for research in this novel space. However, the differential growth trajectories observed across disciplines warrant closer examination. Indeed, despite EJ being an inherently interdisciplinary field, Study 1 showed how EJ research remained siloed within disciplines, highlighting a lack of cross-disciplinary research opportunities. Current EJ literature acknowledges this opportunity gap as a product of established research paradigm and teaching pedagogies (Byrnes et al., 2022). In future interdisciplinary EJ work, Earth science researchers can bring expertise in measuring and explaining physical geologic and environmental processes, while Social science researchers can bring tools for analyzing inequality, community experience, and policy response.

In particular, the Earth science publication group showed a stagnant growth trend in EJ publications, justifying further study to understand the possible roadblocks preventing growth seen by comparable fields in EJ space. While Study 1 identified a clear upward trend, the results represent a snapshot of prior published literature. Future publication patterns may change in response to shifts in federal policy and funding sources. Study 2 aimed to provide clarity through a field-wide survey gauging attitudes and perceptions of EJ by current Earth scientists.

3. Study 2: A Survey on EJ Attitudes and Perceptions Within the Earth Sciences

3.1. Introduction

Study 1 found that the Earth sciences lag behind other major publication fields in producing EJ scholarship. Comparative fields in the Scopus Database, like Environmental Science, contribute significantly more to the exponential growth of EJ publications and, as a result, are keeping up with intellectual demand for novel transdisciplinary work. To better understand the factors contributing to this production gap, Study 2 developed a survey examining three potential barriers to EJ research production among Earth scientists: EJ Knowledge, Implementation Difficulty, and Funding Difficulty. These three dimensions were selected based on previous work emphasizing how current Earth science curricula, research structure and funding opportunities inhibit transformative engagement with EJ (Gomby, 2026; Metzger, 2024), and reflect three supply-side conditions necessary for EJ scholarship: researchers must possess the relevant knowledge, be able to engage and collaborate meaningfully with affected communities, and have access to funding that supports such work. They also expand upon Study 1's takeaways in the context of funding and topic exposure within the field. These dimensions are consistent with broader work showing that environmental health research depends on transdisciplinary collaboration, research infrastructure, and policy support (Biswas et al., 2024), while environmental issues are shaped by broader political conditions that may affect public and institutional support (Birch, 2020). Together, these barriers can help explain why, even when interest exists in the broader academic world, the production of EJ scholarship remains limited in the Earth sciences.

3.2. EJ Knowledge

To properly engage in EJ work, Earth scientists need to know how to structure successful transdisciplinary projects, understand the dynamics of interacting with community stakeholders, and be able to tailor their expertise to specified localized problems (Lang et al., 2012; Luthe, 2017). Developing these skills requires targeted training that integrates EJ principles into Earth science education and research practice. However, the Earth science curriculum often lacks such integrative approaches. Introductory courses often stick to conventional lesson plans that emphasize the technical importance of topics like tectonic evolution, earth surface processes, and geologic formations, often failing to make a deeper societal connection (Egger, 2019). While these topics are foundational for field understanding, the heavy focus on technical content narrows students' exposure to the human dimension of Earth science. The resulting curriculum may alienate students seeking careers with more societal relevance and limit opportunities to connect Earth science with pressing equity and justice issues. As a result, the Earth science introductory curriculum primarily attracts students interested in traditional technical topics and trains them to be the next generation of researchers and teachers. This perpetuates the field's continued lack of skill and interest in addressing societal impact and, by extension, EJ (Orion, 2019). Previous studies have already postulated that an increase in societally relevant and altruistic curricula can increase and retain a diverse student body, providing pathways to explore how the Earth sciences can contribute to conversations on equity and community health

through their disciplinary lens (Moss et al., 2025; Núñez et al., 2020). Although technical competencies could be integrated with EJ through existing frameworks (Jones et al., 2025), it remains unclear whether such integration is occurring across the field en masse. If Earth scientists are not receiving adequate EJ-related training, this knowledge gap may help explain the low EJ publication levels observed in Study 1. Study 2 therefore examined how informed Earth scientists feel about EJ and assessed their familiarity with core EJ concepts.

3.3. EJ Implementation Difficulty

On top of EJ Knowledge, a second challenge arises in understanding the proper process of executing effective EJ-related research in an ideal and ethical manner. Even if Earth scientists possess a substantial knowledge base, applying that knowledge in practice presents additional challenges. These may stem from institutional barriers such as limited administrative support, funding constraints, insufficient capacity, or the exclusion of community agency in project design and implementation (Harrison, 2015).

The goal of EJ work is often to enact change in communities and raise awareness. One common approach is community-based participatory research (CBPR), which emphasizes equal partnerships between researchers and community members in designing and carrying out research projects. However, few frameworks, including CBPR, have presented an effective roadmap to execute these partnerships successfully and address systemic challenges of successful implementation in the higher education curricula (Luthe, 2017; Williamson, 2022). It is more likely that Earth scientists have engaged with community environmental issues through more discrete instances, like one-off events or citizen science engagement workshops. Because Earth scientists' exposure to EJ frameworks may vary widely, Study 2 directly asked participants whether they believe implementing EJ into their research is difficult. Perceived difficulty may reflect two distinct challenges: a lack of foundational EJ knowledge (see Section 3.2), or a deeper barrier related to the absence of clear, actionable frameworks for implementing EJ in scientific research.

3.4. EJ Funding Difficulty

The final notable point of difficulty in developing EJ-focused projects is funding availability and accessibility, specifically for grants dedicated to promoting EJ work (Buchanan & Wozniak-Brown, 2023). In 2021, the United States Environmental Protection Agency (EPA) dedicated only 6 million of its annual 4 billion dollar awarding funds toward EJ-specific endeavors, a number that was shrinking due to the polarized views of the topic (Environmental Protection Agency, 2025a, 2025b; Federal Facilities Environmental Stewardship and Compliance Assistance Center, 2021). This scarcity of targeted funding is not uncommon for emerging or interdisciplinary fields, which are often considered “high-risk” by major funding bodies. At the National Institutes of Health, for example, less than 5% of total funding supports high-risk projects, which includes many EJ-related efforts (Wise et al., 2017). Similarly, the National Science Foundation (NSF) classifies transdisciplinary EJ proposals as high-risk endeavors (National Science Foundation, 2024).

Within the Earth sciences specifically, there is no systematic effort to quantify the number or scope of EJ-focused funding sources. Given the lack of systematic funding data, this study used Earth scientists' perceptions of funding difficulty as an indirect indicator of accessibility. If a large proportion of respondents report high perceived funding difficulty, this may point to funding access as a key structural barrier contributing to the EJ production gap observed in Study 1.

3.5. How Are Demographics Associated With Perceptions of EJ Barriers?

While Earth scientists as a whole might be collectively experiencing these roadblocks toward EJ engagement, the frequency and intensity with which these barriers are encountered are likely to vary across individual demographic characteristics. Previous studies have indicated that demographic factors such as race, gender, and institutional context are not independent of research activities. Rather, they shape research production and teaching in the US (Bratman & DeLince, 2022; Lockhart et al., 2025). Previous work indicated that a researcher's positionality, which includes the researcher's demographic identities, has an influence on EJ research production (Davis & Ramírez-Andreotta, 2021). Race/Ethnicity, for one, has been found to have a direct influence on perception of EJ (Lazri & Konisky, 2019), where people of color have been found to express significantly more concern over injustice within environmental spaces than their white counterparts. These concerns are often grounded in personal and local experiences, such as industrial and transportation-related air pollution in their

communities (Macias, 2016). Taken together, this line of research suggests greater engagement with EJ-related work among some minority groups in Earth sciences.

Gender is also expected to show significant influence on perceptions of EJ, given consistent findings that women tend to express greater concern over pollution, preservation, and resource conservation, amongst other environmental issues (Bell, 2016; Mohai, 1997). Women's heightened concern may make them more attuned to the social and health impacts of environmental inequality, key themes within EJ. Within the Earth sciences, a significant gender bias has been documented in funding biases, resource allocation and workload capacity (Popp et al., 2019), a possible result that can appear within Study 2's measures.

Respondents' proximity to urban-rural centers is expected to emphasize the difference in resource awareness and engagement between urban-rural respondents due to the sheer number of available programs and grants being directed toward urban centers (Brinkley & Wagner, 2024; Temper et al., 2018). As of 2021, California alone has allocated 6.7 billion dollars toward EJ issues within every major metro center in the state (Faust et al., 2021). Study 2's respondents from urban centers are expected to be significantly more involved, aware, and engaged with EJ as a result. Previous work has also highlighted the neglect of EJ-related work in rural areas (Kelly-Reif & Wing, 2016), which may also be reflected in our Earth science respondents' report of lower EJ knowledge in rural areas, along with possible lack of funding for rural EJ projects.

EJ perception can also be influenced by age and career stage. Study 2 focused on respondents' career stage rather than age to better capture their relative experience and EJ exposure within the field. Although career stage is not identical to age, younger respondents are generally expected to be early-career while older respondents are generally expected to be late-career. Earth science students have been shown to value altruistic work in the Earth science field (Carter et al., 2021). However, some foundational Earth science curricula continue to emphasize physical processes and technical training more than the social and community contexts in which environmental problems occur (Jones et al., 2025). Study 2 is expected to show that early-career respondents are more knowledgeable about EJ, partly because younger students have been found to hold a more positive outlook toward societally relevant research and a stronger desire to challenge existing institutional norms (Mayes & Arya, 2024; Moss et al., 2025). It is also expected that later-career respondents show some interest in EJ due to the pervasive nature of the topic, but are less informed on fundamentals and practical EJ methods like co-production of research due to the lack of institutionalized frameworks available in the Earth sciences (Jones et al., 2025; Wyborn et al., 2019).

3.6. Materials and Methods

3.6.1. Data Procurement

Target and snowball sampling approaches on a self-identified population of Earth scientists were used to procure the respondent pool. With protocol approval from the Stanford Institutional Review Board (eProtocol:70205), the survey was developed as a Likert scale questionnaire and was disseminated through leading Earth science society forums, including the American Geophysical Union (AGU), whose membership composition is 55% over the age of 30, over 66% male-identifying, and 70% white according to a 2022 organizational report (American Geophysical Union, 2022) and the Geological Society of America (GSA), whose membership is 72% over the age of 30, 50%–70% male depending on career stage, and over 70% white (Geological Society of America, 2022). Combined membership amounts to upwards of 80,000 Earth scientists worldwide. The forum posts were published and further propagated through weekly society emails, where respondents were encouraged to forward the study to other Earth scientists in their networks. The survey was also shared in Earth science email listservs by community members. The forum post and email text were identical in their body text, where the purpose, incentive, and link were provided to the survey administrator, Qualtrics.

This study also used an in-person recruitment strategy that utilized the annual conferences held by AGU, known as their “Fall Meeting” and GSA's conference “GSA Connects.” The first author attended both as a presenter using an oral format for “GSA Connects” and a Poster format for “Fall Meeting.” The presentation for both these conferences contained motivation for the survey and preliminary results from Study 1. A QR code was distributed at both conferences and could be shared within participants' networks. The survey was conducted from October 2023 to December 2023. The author provided a raffle incentive consisting of \$25 Visa gift cards, which were distributed to randomly selected participants among the survey respondents.

Table 2
Perceptions of Environmental Justice Knowledge

Environmental justice knowledge			
Question	Demographic	Test statistics	Effect size
How informed do you feel about the topic of environmental justice?	Race/ethnicity	1.30	0.18
	Gender identity	0.77	0.10
	City class	0.80	0.11
	Career stage	0.14	0.001
How familiar are you with the concept of community co-production of knowledge?	Race/ethnicity	−0.54	−0.08
	Gender identity	1.40	0.17
	City class	−1.0	−0.14
	Career stage	3.4*	0.02

Note. + $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$. *T*-tests were conducted to compare all demographic groupings except for career stage, where ANOVA significance testing was used. Cohen's *d* was used to measure the effect size for all demographic groupings except for career stage, where η^2 was calculated, and the degrees of freedom for the ANOVA testing were the number of groups − 1 (2).

3.6.2. Participant Population

Responses were collected from 460 self-identified Earth scientists, of whom 369 respondents completed the survey and were included in the analysis. Most of these respondents lived in the United States (75%), with the remaining residing elsewhere (25%). Most participants were affiliated with academic institutions (72%), with the rest working in government (13%), non-profit (4%), industry (6%), or identifying as retired or affiliated with other professional sectors (5%). Two hundred sixty-nine respondents provided complete data on race/ethnicity (76% white; 23% identifying as a minority person of color), 367 on career stage (58% early-career; 25% mid-career, and 17% late-career), 308 on city classification (79% suburban; 21% urban), and 273 on gender identity (58% female, 36% male, 4% non-binary or other gender non-conforming option).

3.6.3. Data Analysis

Data analyses were conducted with R version 4.3.2. We examined how demographics were associated with perceptions of EJ barriers with ANOVA and *t*-tests. Effect sizes were calculated with the “effectsize” package in R (Ben-Shachar et al., 2020).

3.7. Results

3.7.1. EJ Knowledge

The survey first examined differences in EJ knowledge across demographic groups by asking respondents to rate both their familiarity with core EJ concepts and how informed they felt about EJ as a general topic. When asked, “How informed do you feel about the topic of EJ?” “Moderately informed” was the most frequently selected response across all demographic groups, suggesting a shared perception of having some but not extensive knowledge of EJ. Gender, race/ethnicity, city class, and career stage did not predict familiarity with EJ. Career stage was the only significant predictor of community co-production of knowledge. Late-career respondents were less familiar with this concept of co-production than their younger peers: $F(2) = 3.40$, $P = 0.036$, $\eta^2 = 0.02$ (Table 2).

3.7.2. EJ Implementation Difficulty

No significant differences were present between the demographic groupings in terms of perceived difficulty in implementing EJ into their work (Table 3). In terms of local community involvement, Career stage was again the only significant predictor. Earth scientists in the late-career stage tended to report more engagement with their local communities' environmental issues than their younger counterparts ($F(2) = 5.40$, $p = 0.005$, $\eta^2 = 0.03$), a possible reflection of the length of established residency time between the two populations.

Table 3
Perceptions of Environmental Justice Implementation Difficulty

EJ implementation difficulty			
Question	Demographic	Test statistics	Effect size
How difficult would it be to incorporate environmental justice into your work?	Race/ethnicity	0.41	0.06
	Gender identity	−1.60	−0.20
	City class	−0.84	−0.12
	Career stage	0.39	0.002
Some people spend time involving themselves with their local community's environmental issues, while others do not spend time on this. How involved are you with your local community's environmental issues?	Race/ethnicity	−0.55	−0.08
	Gender identity	−0.30	−0.04
	City class	1.60	0.22
	Career stage	5.40**	0.03

Note. + $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$. *T*-tests were conducted to compare all demographic groupings except for career stage, where ANOVA significance testing was used. Cohen's *d* was used to measure the effect size for all demographic groupings except for career stage, where η^2 was calculated, and the degrees of freedom for the ANOVA testing were the number of groups −1 (2).

3.7.3. EJ Funding Difficulty

Finally, we examined whether demographic groupings differed in their levels of perceived funding difficulty with EJ projects. Respondents were asked about EJ funding difficulty and workplace resource awareness to gauge overall EJ funding difficulty.

White respondents perceived an easier time funding projects that included EJ in the Earth sciences than their non-White counterparts ($t = 2.70$, $p = 0.009$, $d = 0.37$). No other demographic grouping yielded a significant result. Gender identity, city class, and career stage did not significantly affect perceptions of funding difficulty for EJ projects. Across these demographic groups, most respondents rated EJ funding as a “slightly difficult” endeavor. This result may indicate that funding could be a barrier to Earth science engagement in EJ, and helps partially explain the production gap found in Study 1. These findings also highlight potential disparities in EJ engagement, specifically between white and non-white Earth scientists.

Results for resource awareness showed a similar pattern where most demographic groupings showed similar responses, responding with some awareness of outreach support, except for city class, which showed urban Earth scientists being more aware of offices or programs geared toward community engagement ($t = 2.50$, $p = 0.015$, $d = 0.34$) (Table 4).

Table 4
Perceptions of Environmental Justice Funding Difficulty

Environmental justice funding difficulty			
Question	Demographic	Test statistics	Effect size
How difficult is it to fund projects that include environmental justice in the Earth sciences?	Race/ethnicity	2.70***	0.37
	Gender identity	0.52	0.07
	City class	−0.68	−0.10
	Career stage	0.29	0.002
To the best of your knowledge, does your workplace have an office or program that provides support for projects that interact with communities, or does your workplace not have such an office or program?	Race/ethnicity	−0.37	−0.05
	Gender identity	−1.40	−0.18
	City class	2.50*	0.34
	Career stage	1.10	0.01

Note. + $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$. *T*-tests were conducted to compare all demographic groupings except for career stage, where ANOVA significance testing was used. Cohen's *d* was used to measure the effect size for all demographic groupings except for career stage, where η^2 was calculated, and the degrees of freedom for the ANOVA testing were the number of groups −1 (2).

3.8. Discussion

Study 2 tested key assumptions related to three potential barriers to EJ engagement: EJ knowledge, implementation difficulty, and funding difficulty. These factors were analyzed across four demographic categories—gender identity, race/ethnicity, career stage, and institutional location (urban vs. suburban)—to examine how perceptions of EJ-related challenges vary within the field. Across the full sample, few statistically significant differences emerged across demographic groups, suggesting that the challenges associated with EJ engagement may be broadly shared rather than confined to specific subgroups.

Career stage was found to influence respondents in more than one measure. Early-career Earth scientists reported greater familiarity with the widely used EJ concept of co-production compared to their late-career counterparts. This is in line with the rise of EJ teachings in recent years, where many instructional frameworks have succeeded in operationalizing EJ principles through community-engaged projects and applied problem-solving exercises (Miles et al., 2025; Trott et al., 2023). Career Stage was also a strong indicator for local community involvement. Late-career Earth scientists were more involved in their local communities' environmental issues than early-career Earth scientists. The correlation between local community engagement and career stage is not surprising because late-career earth scientists were afforded more of an opportunity to establish themselves in their home region, giving them more time to build community (Au et al., 2020). Older populations tend to gravitate toward environmental stewardship for a variety of reasons, including pro-social goals and to combat loneliness (Sundström et al., 2021; Wang et al., 2021). However, age is not the sole determining factor for local community engagement, with lifestyle also impacting volunteering rates significantly (Okun & Michel, 2006). Many late-career scientists are approaching retirement and may have more flexibility to participate in community initiatives. While this involvement is promising, our findings suggest that their participation may be motivated more by general environmental concern or altruism than by an explicit engagement with EJ frameworks, given the current research also found that late-career Earth scientists are less familiar with key EJ concepts.

When examining perceived funding difficulty, race and ethnicity provided greater explanatory insight than career stage. White Earth scientists found it easier to secure funding for EJ-related projects compared to their non-White counterparts. This is not surprising since it has been well documented that white P.I.s and managers receive more funding and favor than their minority peers, typically a result of disproportionate resource allocation and established career attention (Chen et al., 2022; Cruz-Castro et al., 2023; Rosette et al., 2008). While minorities tend to have access to certain targeted programs to promote their engagement (Enriquez et al., 2014), especially in academics through programs like Earth science REUs (Teranes & Kohne, 2013), the difficulty in funding new intersectional projects in the EJ sphere has yet to be fully understood or measured. Given the field's predominantly white, male, and older demographic composition, it is plausible that these intersecting identities collectively shape perceptions of funding accessibility. However, Study 2 did not yield statistically significant results for any demographic factor other than race and ethnicity. We conducted an exploratory model including race/ethnicity, gender, career stage, and their interaction terms to examine whether the race/ethnicity finding may be tied to gender or career stage. The model suggested some interaction patterns, but several subgroup sizes were very small (e.g., only one late-career woman of color). We therefore do not draw definitive conclusions from these interactions. Future interviews or focus groups with Earth scientists across race/ethnicity, gender, and career stage could examine whether perceived EJ funding barriers reflect grant experience, mentoring, institutional support, professional networks, or concerns about the legitimacy of EJ within Earth sciences.

City Class also revealed disparities in perceived resource availability, with urban Earth scientists perceiving more resources for EJ engagement. This result is consistent with previous studies documenting the high volume of EJ work concentrated in urban metropolitan centers, where historic patterns of environmental injustice are prevalent (Anguelovski, 2013; Caniglia et al., 2016; Carvalho et al., 2022). While EJ-related work in non-urban settings has been growing (Brinkley & Wagner, 2024), more work needs to be done to understand to what extent Earth scientists have participated in this shift and whether their community engagement has kept pace with the proliferation of EJ research opportunities in these settings.

EJ engagement in the Earth sciences may also be shaped by broader disciplinary norms about what counts as rigorous or central research. EJ scholarship has been more common in Environmental and Social science fields than in the Earth sciences. Some Earth scientists may view EJ as outside the technical core of their field or as closer to what is sometimes labeled “soft science” (Shapin, 2022). This perception may limit training, mentoring,

publication incentives, and funding support for EJ work, even when Earth science expertise is directly relevant to environmental inequities.

While the survey intended to reach a wide breadth of Earth scientists, its respondent pool is limited to those who opted to complete the survey. Future research can expand on these results by including adjacent Earth science groups like the National Association of Geoscience Teachers (NAGT) and other teaching-focused organizations in the field to capture a broader sample of career Earth scientists.

4. Conclusion

This paper assesses Earth scientists' engagement with EJ through both publication trend analysis (Study 1) and a survey of perceived barriers (Study 2). Study 1 found that Earth science ranked seventh among the top 10 disciplines contributing to EJ scholarship. In contrast, Environmental science and Social sciences were the top two contributors by sheer output and percentage of EJ publications. Despite the Earth sciences' theoretical relevance to EJ, their low output raises questions about barriers to engagement. Study 2 investigated this discrepancy by surveying Earth scientists about perceived challenges to integrating EJ into their work.

Study 2 explored the stagnant growth in Earth science EJ engagement and identified both structural and individual-level barriers, including challenges related to funding and career experience. Although several demographics were taken into account in the analysis, no single demographic factor emerged as a dominant explanation for the lack of EJ engagement. The survey did uncover patterns in familiarity with core EJ concepts, engagement with local communities, and perceived funding difficulty that point to structural and cultural challenges limiting participation. Notably, late-career Earth scientists were more engaged with local communities but less familiar with EJ terminology, suggesting opportunities for collaboration with early-career researchers who are more attuned to EJ discourse. Despite these differences, the field showed general alignment on several EJ-related measures, with a few key areas of divergence.

In all, these findings establish an empirical foundation for understanding certain roadblocks that explain why Earth sciences lag behind in EJ scholarship. We offer a starting point for a broader discussion rather than as a complete explanation of the barriers to EJ engagement in the Earth sciences. As the field of EJ grows, Earth scientists could play a larger role, but doing so will require greater clarity on the institutional, curricular, and professional factors that limit engagement. The present research outlines areas where Earth science training and research culture may be out of step with EJ goals. Future studies, including focus groups or interviews with Earth scientists, could build on these findings to examine how researchers understand these barriers in practice, including limits related to training, funding, community engagement, and disciplinary norms. Such work would help clarify how the Earth sciences can contribute more fully to EJ scholarship and to research on environmental equity and community health.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

Supporting bibliometric and survey data are available on the associated Open Science Framework (OSF) page (Rosales Cortez & Chen, 2026). Files include the raw Scopus CSV files used for the Study 1 bibliometric analysis and de-identified survey data from Study 2. The associated code used to analyze data from both studies is also available on the OSF page (<https://doi.org/10.17605/OSF.IO/D7ZA6>).

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